APL — Acute Promyelocytic Leukemia Emergency & Risk-Adapted Therapy

1. Sinking ship / DIC

WHAT YOU SEE

A sinking ship bleeding red into the water, crew labeled 'GI / BRAIN / LUNG'.

WHAT IT ENCODES

APL (t(15;17), PML-RARA) presents with life-threatening coagulopathy: DIC plus hyperfibrinolysis causing GI, intracranial, and pulmonary hemorrhage. Early hemorrhagic death is the main killer.

BOARD TRAP

The lethal early event in APL is bleeding from DIC, not infection or leukostasis. Aggressive blood-product support and prompt ATRA save lives.

2. APL ship label

WHAT YOU SEE

'APL' on the ship hull.

WHAT IT ENCODES

Acute promyelocytic leukemia = AML M3, defined by t(15;17) PML-RARA fusion. Promyelocytes packed with Auer rods (faggot cells). The fusion blocks myeloid differentiation at the promyelocyte stage.

BOARD TRAP

Confirm PML-RARA by PCR/FISH; variant translocations (e.g. t(11;17) PLZF-RARA) are ATRA-resistant. But never delay ATRA awaiting confirmation.

3. ATRA — start on suspicion

WHAT YOU SEE

Differentiation theme; figures transforming on the dock.

WHAT IT ENCODES

All-trans retinoic acid (ATRA) overcomes the differentiation block, maturing promyelocytes into neutrophils. Start ATRA the moment APL is suspected — before genetic confirmation — to reduce fatal hemorrhage.

BOARD TRAP

Do NOT wait for cytogenetics to start ATRA. Delaying differentiation therapy in suspected APL is the classic fatal mistake.

4. 85% cure

WHAT YOU SEE

'85%' plaque on the safe sunny dock.

WHAT IT ENCODES

APL is the most curable AML: ATRA + arsenic trioxide (ATO) achieves ~85–90%+ cure in low/intermediate risk, largely chemo-free.

BOARD TRAP

Despite the best prognosis of any AML, untreated APL has the highest early mortality from DIC — favorable long-term but emergent up front.

5. WBC risk cutoff 10,000

WHAT YOU SEE

Gauge 'LOW RISK ≤10,000' vs 'HIGH RISK >10,000' WBC.

WHAT IT ENCODES

Sanz risk stratification uses presenting WBC: ≤10,000 = low/intermediate risk; >10,000 = high risk. WBC drives both treatment intensity and differentiation-syndrome risk.

BOARD TRAP

The APL risk cutoff is WBC 10,000 (with platelet count subdividing low vs intermediate). High WBC is the key high-risk discriminator.

6. Low risk — no chemo

WHAT YOU SEE

'NO CHEMO' gate on the low-risk path.

WHAT IT ENCODES

Low/intermediate-risk APL is treated with ATRA + arsenic trioxide alone — a chemotherapy-free regimen with excellent cure rates and less toxicity.

BOARD TRAP

Low-risk APL does not need anthracycline; ATRA+ATO suffices. Adding cytotoxic chemo here is unnecessary toxicity.

7. High risk — add chemo

WHAT YOU SEE

'HIGH RISK' gate, anthracycline figure.

WHAT IT ENCODES

High-risk APL (WBC >10,000) adds an anthracycline (e.g. idarubicin) or gemtuzumab to ATRA+ATO for cytoreduction and CNS-protective benefit.

BOARD TRAP

High WBC APL needs added cytoreduction; treating high-risk disease with ATRA+ATO alone undertreats it.

8. APL differentiation syndrome

WHAT YOU SEE

'APL SYNDROME 1-3 WEEKS' warning sign.

WHAT IT ENCODES

Differentiation (retinoic acid) syndrome: fever, dyspnea, pulmonary infiltrates, weight gain, effusions, hypotension, AKI — typically 1–3 weeks into ATRA/ATO. Promptly give dexamethasone.

BOARD TRAP

Differentiation syndrome is treated with dexamethasone 10 mg BID and supportive care; ATRA is held only if severe. Don't mistake the infiltrates for pneumonia/fluid overload alone.

9. Differentiation Syndrome Bay

WHAT YOU SEE

'DIFFERENTIATION SYNDROME BAY' with chest X-rays and 'DEX 10 Q12H'.

WHAT IT ENCODES

Management bay: chest imaging shows bilateral infiltrates; treat with dexamethasone 10 mg every 12 hours. Leukocytosis worsens risk, so high-WBC patients are watched closely.

BOARD TRAP

Bilateral infiltrates + ATRA = think differentiation syndrome, not simply infection — empiric dexamethasone is the move while covering broadly.

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10. 10% mortality

WHAT YOU SEE

'10% MORTALITY' sign in the syndrome bay.

WHAT IT ENCODES

Early death (mainly hemorrhage, also differentiation syndrome) accounts for roughly the residual mortality despite high curability — concentrated in the first weeks.

BOARD TRAP

APL mortality is front-loaded; survive the early DIC/differentiation phase and long-term outcomes are excellent.

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11. Blood product support

WHAT YOU SEE

'FFP / CRYO / PLT / PRBC' transfusion bags.

WHAT IT ENCODES

Aggressive coagulopathy support: keep fibrinogen >100–150 with cryoprecipitate, correct PT/PTT with FFP, transfuse platelets to keep counts high (often >30–50k) given bleeding risk.

BOARD TRAP

In APL, platelets are kept much higher than the usual 10k threshold because of DIC bleeding risk — undertransfusing here is dangerous.

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12. PCR / MRD monitoring

WHAT YOU SEE

'PCR' device, green buoy, clinician.

WHAT IT ENCODES

PML-RARA transcript by RT-PCR tracks MRD; molecular remission is the goal, and rising transcripts signal relapse. Guides duration of therapy and detects early relapse.

BOARD TRAP

APL response is judged molecularly (PML-RARA PCR), not just morphologically — a morphologic CR with persistent PCR positivity is not adequate.
